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Text Sentiment Analyser: A Hybrid Rule-Based and Machine Learning Framework for Sentiment and Emotion Analysis of English and Code-Mixed Hinglish Text

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Abstract. Indian social-media users often type Hindi in the Latin alphabet and switch to English within one sentence, a style known as Hinglish. English sentiment tools handle such text poorly because Romanised Hindi opinion words are missing from their lexicons, spellings vary widely and Hindi places the negator after the word it negates. This paper presents Text Sentiment Analyser, a privacy-preserving system that labels English and Hinglish text as Positive, Neutral or Negative and profiles its emotions. The design joins four parts: a language identifier with a spelling normaliser; a readable bilingual word list whose rules cover negation placed before or after a word, words such as *bahut* that strengthen an opinion, and contrast words such as *lekin*; two trained classifiers (multinomial Naive Bayes and logistic regression on TF-IDF features of single words and word pairs); and a weighted blend of the rule scores with Naive Bayes. Extra modules report Plutchik's eight emotions, opinions about named product aspects, possible sarcasm and the words that drove each decision. We tested five engines on 210 balanced, hand-labelled texts using five-fold stratified cross-validation. Accuracy was 55.2% for an English-only word list, 92.4% for our Hinglish-aware rules (95% CI 88.6–95.7), 65.9% for Naive Bayes and 82.1% for the blend, averaged over five random splits; the blend beat Naive Bayes significantly (McNemar $p < 0.001$). Since our word list and test texts were prepared together, the 92.4% figure is best treated as a ceiling. Ablation, learning-curve and error studies show which parts matter. The tool works offline inside a browser.

Keywords: *Hinglish, Code-mixing, Opinion mining, Rule-based classifier, Naive Bayes, Emotion recognition, Explainable NLP, Text classification*

Introduction:

A. Background and Motivation

Sentiment analysis is the computational study of the opinions and feelings people express in text, and it supports review mining, brand monitoring, complaint triage and opinion research [1–3]. Its best-known tools, including VADER [4], SO-CAL [5] and the supervised classifiers of Pang et al. [6], were built for English. Many Indian users, however, switch between Hindi and English within a sentence and type Hindi with a Roman keyboard [7], which gives comments like *"movie acchi nahi thi but songs mast the"*, meaning that the film disappointed but its songs were good.

Such text defeats English tools in three ways. Hindi opinion words such as *bakwas* (rubbish) and *zabardast* (excellent) are absent from English lexicons; Romanised Hindi has no standard spelling, so *bahut* (very) also appears as *bahot*, *bhot* or *bohot*; and Hindi places the negator after the word it negates (*acchi nahi thi*), so English negation windows point the wrong way. Code-mixing is known to harm language identification, tagging and sentiment classification [8–10].

B. Problem Statement and Objectives

Tools built for English mark a large share of Hinglish comments as Neutral, and trained models would need labelled Hinglish examples, which a student project seldom has in quantity. We therefore set out (i) to build a rule engine that understands Hinglish spellings, negation in both languages, intensifying words and contrast words; (ii) to train Naive Bayes and logistic regression on labelled examples; (iii) to blend the rules with Naive Bayes and find out when blending pays off; (iv) to report emotions, aspects, sarcasm hints and word-level reasons; and (v) to evaluate all engines with reproducible statistics. Our contributions are a pipeline with no outside libraries whose resources hold 292 English and 184 Hinglish opinion words, 28 Hindi intensifiers and 94 spelling-variant pairs, a five-engine comparison with confidence intervals and significance tests, and ablation, learning-curve and error analyses.

Literature Review:

[3] Birjali et al. (2021) and [11] Wankhade et al. (2022) surveyed lexicon-based, machine-learning and hybrid sentiment methods and identified negation, sarcasm and code-mixed, low-resource text as open problems. [12] Nandwani and Verma (2021) reviewed sentiment analysis together with emotion detection, and [13] Bordoloi and Biswas (2023) proposed a design framework, from pre-processing to visualisation, that matches the pipeline used here.

[14] Medhat et al. (2014) and [15] Zhang et al. (2018) reviewed classical and deep sentiment algorithms. Later reviews by [16] Ligthart et al. (2021), [17] Jain et al. (2021), [18] Tan et al. (2023), [19] Rodríguez-Ibánez et al. (2023) and [20] Jim et al. (2024) traced the move towards transformers, yet all of them still list informal, emoji-rich and code-switched text as unsolved. [21] Hartmann et al. (2023) compared methods on 272 marketing datasets and found transfer learning the most accurate, with lexicons faster and easier to inspect.

[4] Hutto and Gilbert (2014) proposed VADER, a validated lexicon with heuristics for punctuation, capitals, degree words, *but* and negation, normalised into a compound score that our engine adopts; [5] Taboada et al. (2011) modelled negation as an attenuating shift of polarity. [22] Catelli et al. (2022) found BERT more accurate than a lexicon on Italian but the lexicon faster, data-free and interpretable. [6] Pang et al. (2002) and [23] Wang and Manning (2012) showed that Naive Bayes and linear models over unigrams and bigrams are strong baselines, and [24] Tan et al. (2022) combined RoBERTa with an LSTM for higher accuracy on large English corpora.

[25] Thelwall et al. (2010) and [26] Kiritchenko et al. (2014) showed that lexicon features work well on short informal text, and [27] Cambria et al. (2022) used commonsense knowledge to make polarity decisions explainable in SenticNet 7. Hybrids that join lexicons with classifiers were proposed by [28] Mudinas et al. (2012) and [29] Appel et al. (2016); more recently [30] Kaur and Sharma (2023) mixed handcrafted with learned features and [31] Alsayat (2022) built an ensemble of language models.

Neural models followed, from convolutional networks [32] to BERT [33]. Comparisons by [34] Dang et al. (2020) and [35] Yadav and Vishwakarma (2020) show that their lead depends on large labelled corpora. Strong transformer results on English reviews and tweets were reported by [36] Alaparthi and Mishra (2021), [37] Kokab et al. (2022) and [38] Bello et al. (2023), and shared resources include TweetEval [39], XLM-T [40] and pysentimiento [41]. None of these covers Romanised Hinglish.

[42] Doğruöz et al. (2021) surveyed code-switching from linguistic and social perspectives. [43] Khanuja et al. (2020) built the GLUECoS benchmark for code-switched NLP, and [44] Nayak and Joshi (2022) pre-trained HingBERT on a large Hinglish corpus. [45] Joshi et al. (2016) introduced a sub-word LSTM for Hindi-English comments that was 4–5% more accurate than traditional methods, [10] Patra et al. (2018) fixed the three-class task in the SAIL shared task, and [46] Patwa et al. (2020) organised SemEval-2020 SentiMix, where top systems such as that of [47] Srinivasan (2020) fine-tuned multilingual transformers [48]. [49] Singh and Lefever (2020) transferred English knowledge through cross-lingual embeddings, whereas [50] Baroi et al. (2020) obtained an F-score of only 0.617 with a classical ensemble.

[51] Sitaram et al. (2019) surveyed code-switched speech and language processing, and [52] Winata et al. (2023) reviewed several decades of code-switching work in NLP. The toolkit has since grown to include the LinCE benchmark [53], studies of how multilingual BERT handles mixed text [54, 55], methods for generating synthetic code-mixed sentences [56, 57], a syntactic measure of code-mixing [58], Hinglish parallel and evaluation corpora [59, 60] and code-mixed translation [61]. Indic pre-trained models such as MuRIL [62] and IndicBERT [63] are obvious baselines for future comparison.

In the SAIL task, [64] Sarkar (2018) came third with multinomial Naive Bayes over n-grams, which supports our choice of baseline. Other SentiMix entries used XLM-R with focal loss [65], a CNN with self-attention [66] and a syntactic-semantic LSTM [67]. Related datasets cover Tamil-English [68], Malayalam-English [69] and Kannada-English [70] sentiment and Hindi-English hate speech [71, 72], and [73] Khan et al. (2022) applied multilingual BERT to Urdu.

[74] Chakravarthi et al. (2022) released DravidianCodeMix and highlighted class imbalance and annotation cost. [75] Mehmood et al. (2020) showed that normalising Roman Hindi and Urdu spellings improves sentiment classification, and [76] Mabokela et al. (2023) concluded that lexicons and normalisation remain vital for under-resourced languages. [77] Plutchik (1980) proposed eight basic emotions, used in the NRC lexicon of [78] Mohammad and Turney (2013). [79] Ribeiro et al. (2016) introduced perturbation-based explanation and [80] Ribeiro et al. (2020) exposed negation failures in commercial sentiment models through behavioural tests. Finally, [81] Zhang et al. (2024) and [82] Zhang et al. (2023) found that large language models lag on fine-grained and code-switched tasks and require network access.

Emotion resources include WordNet-Affect [83] and GoEmotions [84], both covered in the review by [85] Acheampong et al. (2020). Sarcasm stays hard for people and machines alike [86, 87], and the polarity of emoji has been measured on a large scale [88]. [89] Danilevsky et al. (2020) surveyed explanation methods for NLP and [90] Atanasova et al. (2020) compared them. ChatGPT-style models were tested on sentiment by [91] Wang et al. (2023) and [92] Kheiri and Karimi (2023), while aspect-level [93] and multimodal [94] analysis have their own recent surveys.

Research gap. Rule-based tools are English-only, strong code-mixed systems need large labelled corpora and GPUs, few systems explain verdicts word by word, and small studies seldom report confidence intervals or ablations. The present work addresses these gaps.

Methods and Materials:

A. Architecture and Corpus

Text enters by keyboard, voice or CSV upload (Fig. 1). A language identifier tags it as English, Roman Hindi, Hinglish or Devanagari, and a normaliser collapses repeated letters and maps spelling variants to canonical forms. The cleaned tokens go to the rule engine and to both classifiers at the same time, after which a blending step returns a verdict with a confidence value; separate modules compute emotions, aspect-level sentiment, a sarcasm indicator, a word-level explanation and text statistics. The system has about 4,300 lines of JavaScript and runs in a browser or Node.js; a Python script reproduces the experiments with scikit-learn, NumPy and Matplotlib [95–97].

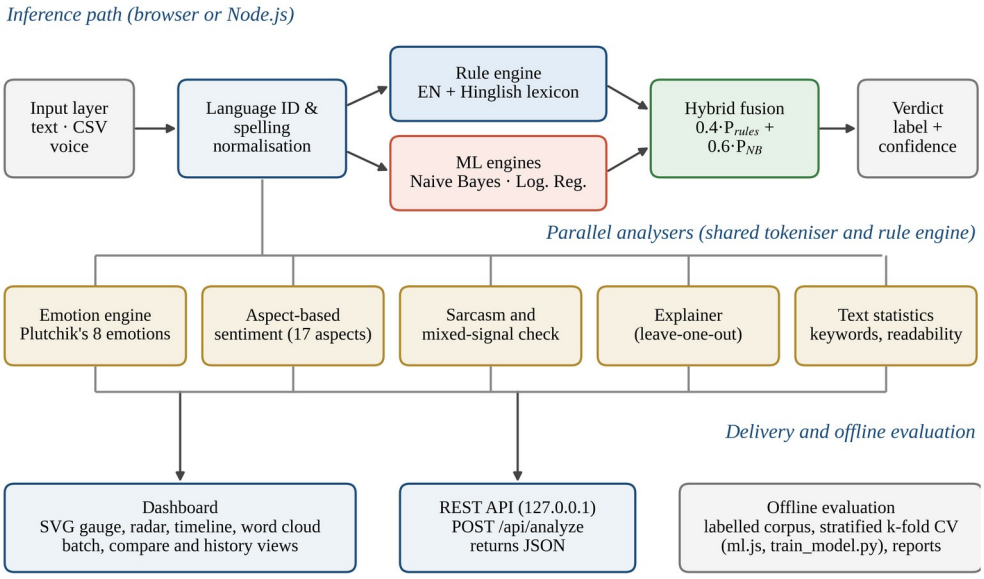


Fig. 1. Architecture of Text Sentiment Analyser: inference path (top), parallel analysers (middle), and delivery and offline evaluation (bottom).

Our test set contains 210 short English and Hinglish sentences that we wrote ourselves, modelled on public comments about films, phones, cricket, food, travel, college and customer service (Positive 71, Neutral 69, Negative 70; mean length 7.1 words; 133 Hinglish, 63 Roman Hindi and 14 English). When a comment mixes opinions we labelled it by its main or closing opinion, sarcastic comments by what the writer really meant, and plain facts or questions as Neutral. Because the Hinglish word list was compiled alongside the corpus, the rule engine's score is an optimistic upper bound; the learned models are always tested on held-out folds.

B. Normalisation and Rule-Based Engine

Following standard NLP practice [98], the language identifier counts tokens from a 211-word Romanised-Hindi list and an English function-word list (Hinglish when the Hindi share reaches 15%). The normaliser lower-cases tokens, reduces runs of three or more identical letters to two (*bahuuut* → *bahuut*) and applies 94 variant mappings (*bhot*, *bohot* → *bahut*; *nhi* → *nahi*).

The rule engine scans tokens once, accumulating positive evidence P and negative evidence N (Table 1). Lexicon words carry valences of 1–4 and emoji −4 to +4. An English negator reverses and attenuates sentiment words in the next three tokens by −0.75 [5]; a Hindi negator instead looks

back up to three words and multiplies the most recent sentiment score by −0.75, so *acchi nahi thi* becomes negative, while emoji are never reversed. Boosters scale the next word (*bahut* ×1.6, *thoda* ×0.6), and a contrast word (*but*, *lekin*, *magar*) halves earlier evidence and multiplies later evidence by 1.5. Eq. (1) gives a token's score and Eq. (2) the compound score [4]; the text is called Positive when c is at least 0.05 with P above N, Negative when c is at most −0.05 with N above P, and Neutral in every other case.

$$s_t = v_t \cdot b_t \cdot \eta_t \cdot \kappa_t \tag{1}$$

$$c = x / \sqrt{(x^2 + 15)}, \quad x = P - N \tag{2}$$

In Eq. (1), v_t is the word's stored strength and b_t its booster multiplier; η_t is −0.75 when the word is negated and 1 if not, and κ_t is 1.5 for words after a contrast word and 1 if not.

Table 1. Main parameters of the rule engine and linguistic resources.

Parameter / resource	Value
English / Hinglish lexicon	292 / 184 entries, valence 1–4; emoji −4 to +4
Negation	Factor −0.75; English forward window and Hindi look-back of 3 tokens
Intensifiers	28 Hindi and 27 English; factors 0.5–1.8
Contrast words (but, lekin, magar)	×0.5 before, ×1.5 after
Emphasis	ALL-CAPS ×1.3; each "!" +8% (at most four)
Emotion lexicon and aspects	376 word–emotion entries (8 emotions); 17 aspect categories

C. Machine Learning and Hybrid Fusion

Both classifiers use normalised unigrams and bigrams, keeping emoji, "!" and "?" as tokens. Multinomial Naive Bayes [99] uses Laplace smoothing (Eq. 3, α = 1). Logistic regression uses sub-linear TF-IDF vectors [100, 101] with a softmax output trained by stochastic gradient descent [102] with L2 regularisation (40 epochs, $\eta_0 = 0.4$, $\lambda = 10^{-4}$). The hybrid converts c into probabilities and mixes them with the Naive Bayes posterior (Eqs. 4–5); the label is the arg-max.

$$P(f \mid y) = (n_{f,y} + \alpha) / (N_y + \alpha|V|) \tag{3}$$

$$P_r(\text{Pos}) = \max(c, 0), \quad P_r(\text{Neg}) = \max(-c, 0), \quad P_r(\text{Neu}) = 0.9 \cdot \max(0, 1 - |c|) + 0.05 \tag{4}$$

$$P_{\text{hybrid}}(y) = w \cdot P_r(y) + (1 - w) \cdot P_{\text{NB}}(y), \quad w = 0.4 \tag{5}$$

The emotion module assigns lexicon evidence to Plutchik's eight emotions [77]; the aspect module scores clauses containing keywords such as *camera* or *battery*; the sarcasm module reports rule-based cues (eye-roll emoji, *wah kya baat hai*); and the explainer measures each word's impact by deleting it and rescoring, $\Delta_i = c(T) - c(T \setminus t_i)$ [79]. All charts are hand-written SVG, a content-security policy blocks outgoing requests, and the optional REST API listens on 127.0.0.1 only.

D. Evaluation Protocol

Engines are evaluated by stratified five-fold cross-validation [103], with the primary split at seed 7 and repeats at seeds 11, 21, 42 and 99. Metrics are accuracy, macro-F1, Cohen's kappa [104] and 95% confidence intervals from 2,000 bootstrap resamples [105]; engines are compared with McNemar's test [106, 107], and kappa is read on the Landis–Koch scale [108]. Ablations remove one rule component at a time, and further runs sweep w and vary the training size. No setting was tuned to improve headline figures.

Results:

A. Overall Performance

The headline figures are in Table 2 and Fig. 2. With English words alone the engine scores 55.2%, mostly because it calls 76% of all texts Neutral. The Hinglish layer raises accuracy to 92.4% (κ = 0.89), a 37.2-point gain. Naive Bayes and logistic regression average 65.9% over five splits, and the hybrid 82.1%, 16.2 points above Naive Bayes. The scikit-learn models behave similarly (Naive Bayes 62.4%, logistic regression 65.7%, linear SVM [109] 64.8%), which suggests that, with roughly 170 sentences to learn from, the shortage of data matters more than which classifier is used.

Table 2. Performance of the five engines (stratified five-fold cross-validation, n = 210; primary split except the last column).

Engine	Accuracy (%)	Macro-F1 (%)	95% CI (%)	κ	Five-split mean (%)
English lexicon (baseline)	55.3 ± 6.4	52.8 ± 8.0	48.6–61.9	0.33	55.2 ± 0.0
Hinglish-aware rules (ours)	92.4 ± 2.4	92.4 ± 2.2	88.6–95.7	0.89	92.4 ± 0.0
Naive Bayes	70.0 ± 3.1	69.9 ± 3.6	63.8–75.7	0.55	65.9 ± 2.8
Logistic regression	67.6 ± 2.8	67.3 ± 2.9	61.4–73.3	0.51	65.9 ± 1.8
Hybrid: rules + NB (ours)	83.8 ± 4.7	83.7 ± 4.6	78.6–88.6	0.76	82.1 ± 1.6

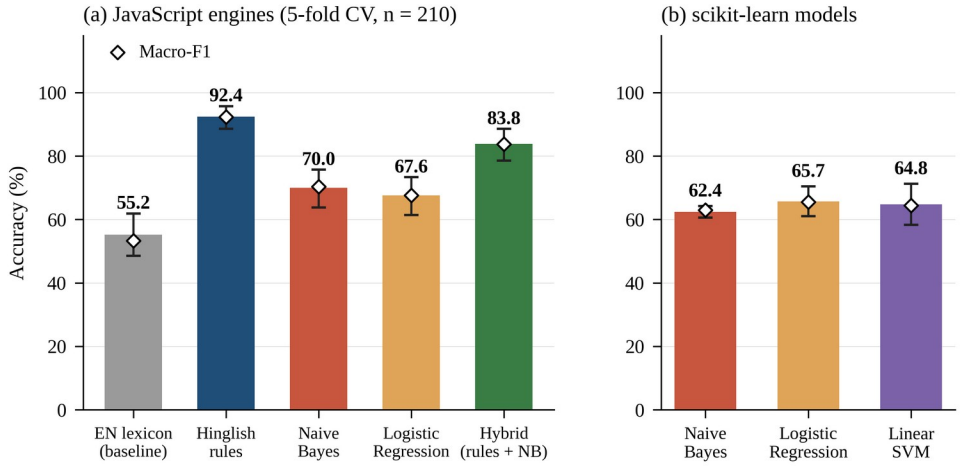


Fig. 2. Accuracy of the compared engines: (a) JavaScript engines with 95% bootstrap intervals; (b) scikit-learn models (mean ± SD over five folds). Diamonds show macro-F1.

B. Statistical Reliability and Class Behaviour

Bootstrap intervals are wide (half-widths 3.6–6.7 points), so only large differences are reliable. Paired McNemar tests show the Hinglish rules ahead of the English baseline (81 texts against 3, $p < 0.0001$), the blend ahead of Naive Bayes (30 against 1) and of logistic regression (35 against 1, both $p < 0.0001$), and the rules ahead of the blend (24 against 6, $p = 0.002$), whereas Naive Bayes and logistic regression cannot be separated ($p = 0.27$). All engines except the baseline recognise Neutral texts well (F1 0.87–0.92), but the learned models confuse Positive and Negative (Fig. 3) because unseen opinion words contribute nothing at test time, while the Hinglish engine mainly misses complaints that contain no lexicon word.

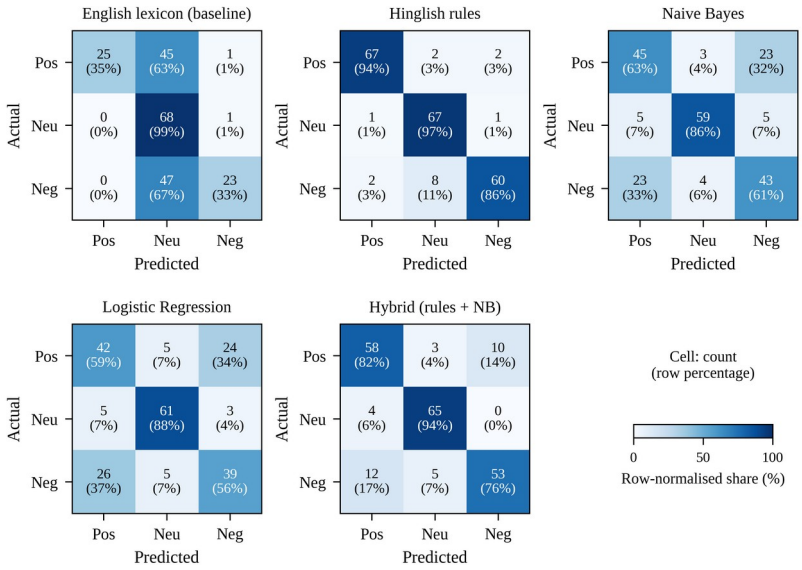


Fig. 3. Confusion matrices of the five engines (pooled out-of-fold predictions; rows are actual classes).

C. Ablation, Sensitivity and Data Efficiency

Removing the Hindi lexicon drops accuracy from 92.4% to 55.2% (Table 3). Taking out any other single rule moves overall accuracy by one point or less, but it does break hand-picked test sentences like *movie acchi nahi thi* and *camera mast hai but battery bekaar*, because the corpus contains few negators and contrast words [80]. Post-negation corrects four texts and breaks four, two of them “never … such” idioms with *kabhi nahi*. Varying w (Fig. 4a) gives a maximum of 89.4% at w = 0.7, an optimistic bound since it is chosen on the test folds. With only 17 training texts, Naive Bayes scores 49.9% but the hybrid already 75.5% (Fig. 4b), showing that the rules supply prior knowledge when data are scarce.

Table 3. Ablation of the Hinglish rule engine (Δ in percentage points; last column: 12 targeted test sentences).

Variant	Accuracy (%)	Δ	Texts changed	Tests passed
Full Hinglish engine	92.4	—	0	12/12
− Hindi post-negation	92.4	0.0	8	9/12
− Contrast rule (but, lekin)	92.4	0.0	0	10/12
− Intensifiers (bahut, ekdum)	91.9	−0.5	1	12/12
− English pre-negation	92.4	0.0	0	10/12
− Emoji and emoticon scores	91.4	−1.0	2	11/12
− Hindi lexicon (English only)	55.2	−37.1	86	5/12

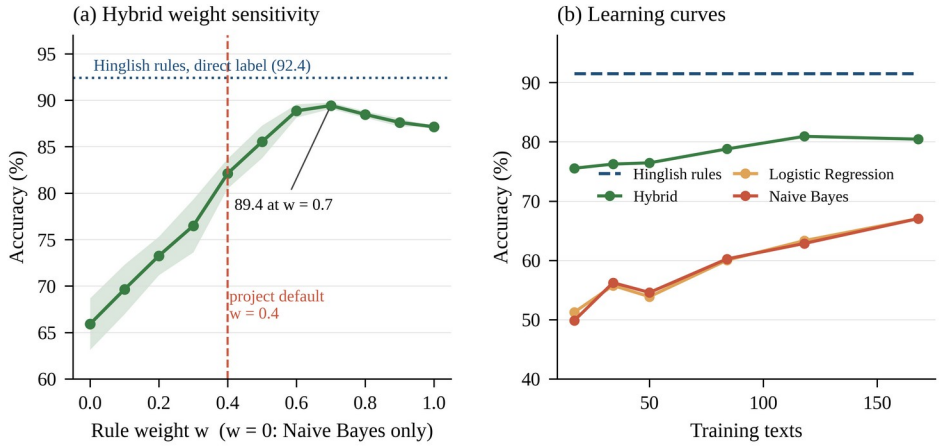


Fig. 4. (a) Hybrid accuracy as the rule weight w varies; (b) learning curves of the learned models, the hybrid and the rule engine.

D. Error Analysis and Runtime

The 16 errors of the Hinglish engine fall into three groups: nine implicit complaints with no lexicon word (*Battery bahut jaldi khatam ho jaati hai*), four negation-scope errors involving *kabhi nahi* or comparatives, and three domain-dependent words (*problem*, *jeet*, *refund*). The learned models fail mainly on unseen vocabulary; Naive Bayes labels *Heroine ki acting lajawab thi* Negative because *lajawab* never occurs in training. On *movie acchi nahi thi but songs mast the* the engine gives +0.70 (Positive), and the explanation view shows *mast* (+0.89) and *but* (+0.34) doing most of the work. All automated tests pass, and one text is analysed in 0.12 ms on average.

Figures and Tables

Note: all figures were produced at 600 dpi with labelled axes, and the first row of every table is a header row.

Discussion:

The central finding is that, when labelled data are scarce, vocabulary coverage rather than algorithm choice decides accuracy on Hinglish: lexicon entries added 37 points, while moving between Naive Bayes, logistic regression and a linear SVM shifted results by just a few points, consistent with earlier observations on linear baselines [23] and under-resourced languages [76]. The hybrid improved clearly on Naive Bayes, especially with little data, but did not beat the rule engine, partly because the lexicon was built with the corpus and partly because Eq. (4) suppresses weak rule signals while Naive Bayes posteriors are extreme. Corpus accuracy also hid the value of rules that fixed targeted cases, so challenge sets should accompany any benchmark [80].

The system is transparent, fast (about 0.1 ms per text) and fully offline, which suits privacy-sensitive uses and makes it a baseline for heavier models, including large language models [81, 82]. Its limitations are a small, team-written and balanced corpus, an optimistic rule-based score, unmeasured inter-annotator agreement, a fixed hybrid weight, unscored emotion and sarcasm modules, and Roman-script-only input. Future work will collect at least 1,000 real doubly annotated comments, tune w by nested validation, and compare against MuRIL-style Indic transformers and HingBERT [44].

Conclusion:

We built Text Sentiment Analyser to read opinions and emotions in English and Hinglish without sending any text off the user's machine, using hand-written language rules alongside trained classifiers. On our 210 test sentences, the Hinglish-aware engine reached 92.4% accuracy against 55.2% for an English-only lexicon. The hybrid reached 82.1%, about 16 points better than Naive Bayes, and this gap grew larger when less training data was available.

We wrote the word list and the test sentences together, so the rule score should be read as a best case, and 210 sentences cannot stand in for real social-media traffic. What the work does offer is a design whose every decision can be traced, an evaluation that others can repeat, and evidence about which parts actually help. Our next step is to test it on real comments labelled by two independent annotators.

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